

Detection of Dental Caries Using Radiographic Images

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TEAMWORK REFLECTIONS

Our team was made up of three students namely Abdulghani Mohammad, Abdulraheem Al-Ani, and Mohamad Kassem. We had a definite separation of responsibilities at the start and still maintain a shared decision-making process on all major issues. Abdulghani concentrated primarily on data preparation (Kaggle cavity), execution of the preprocessing pipeline (grayscale conversion, resizing, CLAHE and normalisation) and writing the methodology and literature review parts of the paper. Abdulraheem led in terms of developing models and experimenting with them, beginning with the standard CNN baseline and then proceeding to transfer learning with VGG16 in both binary classification and 3-class severity classification. Mohamad focused on the development of the Streamlit user interface, which produces the visual illustrations of the system predictions and assists in the format and figures of IEEE-style report. They communicated via a WhatsApp group and brief face-to-face communication in the lab mostly. It was a good blend of both styles: online messages were sufficient to answer the questions fast, whereas the face-to-face meetings were applied to address the more complicated issues like the division of the dataset, determination of the severity, and the final selection of the hyperparameters. In cases where one of the members was faced with a coding issue, particularly shape mismatches in Tensorflow or path errors in Colab, the rest of the group would tend to have a joint notebooks session and debug together. This not only hastened the speed, but also guaranteed that no one was unaware of the entire pipeline end-to-end. Organizational problems were rather than technical the chief ones. Since we tried a lot of different model variants in Colab, it was not hard to lose track of which run resulted in what outcome. To overcome this we finally fully standardised model and weight file names and stored end-result metrics in a comparison table included in the report. The other challenge was the interpretation of "early" vs. "advanced" caries based on the definition of the labels of the bounding-box. Having tried a number of experiments and deliberations, we finally agreed on a simple rule of threshold which created fairly balanced classes. The overall experience in the teamwork was good and every individual member played an important role as per his strengths.

IMPACT ANALYSIS

Technically, the project indicates that a relatively simplistic transfer-learning system, which involves VGG16, and contrast enhancement by CLAHE, can achieve believable results when it comes to detecting binary caries and grading their severity on intra-oral images. The three-class model was trained on a small sample, but it can distinguish between healthy and early lesions and more severe decay with a reasonable level of accuracy. This indicates that with bigger and more representative clinical data sets, similar architectures can be the source of pragmatic decision-support solutions in dental care. Such system would not substitute a dentist, but may be used as a second reader of radiographic images. An AI model that has been embedded into a radiography workflow in a clinic with high traffic would be able to automatically identify suspicious images, cases that have a high likelihood of an early or advanced caries, and offer a quantitative assessment of confidence. This could help in increase early diagnosis rates, decrease the rate of missed lesions and standardise diagnostic quality among the practitioners of various levels of experience. Our Streamlit interface is just a prototype, but it shows how these concepts might look to a clinician in such a way that it is easy to understand and use. Meanwhile, the weaknesses of the existing work are also obvious. The sample is quite small and might not represent the entire range of real clinical radiographs. The number of bounding boxes is used to come up with our severity labels instead of being graded by experts, which adds noise to the ground truth. The system has further not been tested in a practical clinical setting, but merely offline. Due to these reasons, we are clear in placing the model as a research prototype, and we highlight this fact in the user interface that definitive diagnosis is always the work of a qualified dentist. However, the project offers a tangible example of dental imaging with the application of deep learning and can be a foundation of more sophisticated work in the future.

NEW LEARNING AND SKILLS GAINED

Technically the project gave us the practical experience of the whole pipeline of deep learning with the help of TensorFlow/Keras. We performed our own convolutional network implementations, loaded and edited already trained backbone of VGG16, froze and unfroze layers and tested with learning

rates, batch sizes and loss functions. In addition to this, we gained practical understanding of image processing such as conversion to grayscale, CLAHE contrast improvement, Gaussian smoothing and normalisation and knew the impact of these alternatives on convergence and generalisation. We also learned how to minimize imbalance in classes using the aid of class weights and the interpretation of different evaluation values, including confusion matrices, precision-recall trade-offs, F1-scores and ROC curves. The non-technical skills were also significant. The necessity to revise more than fifty papers to lead a literature review made us critical in reading them, comparing datasets and architectures, and identifying real research gaps instead of presenting the list of results. We managed to improve our academic writing and reference skills because of the format of writing the report in the IEEE conference. Impression implementation and debugging of the Streamlit user interface also led to the introduction to basics of model deployment and user-centred design. Finally, the process of the work organization through the Gantt chart and weekly meetings also provided me with some experience in managing project, planning time, and solving problems in the team. Overall, the project both broadened our background knowledge on deep learning in medical imaging, and provided us with teamwork-based research and engineering projects to do in the future.